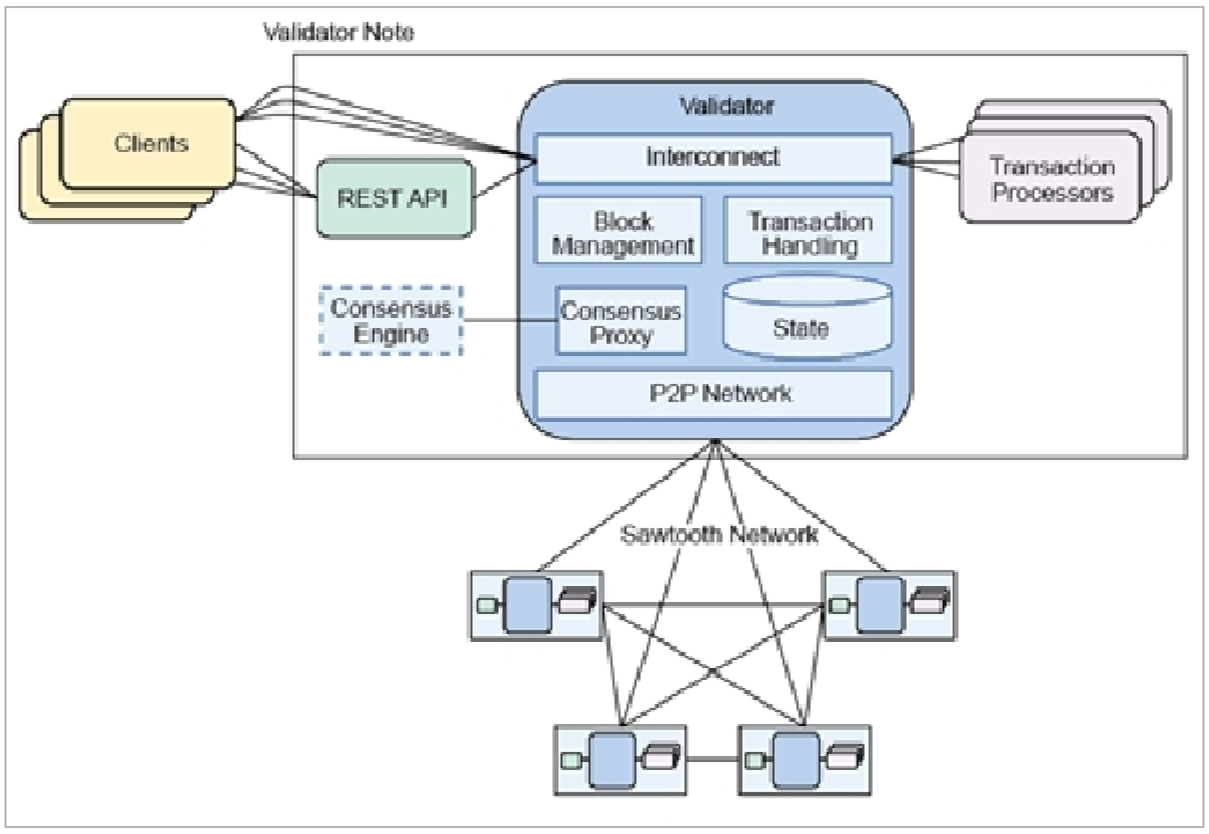

Figure 4: Hyperledger Sawtooth architecture (source: *https://sawtooth.hyperledger.org)*

and decided by the use case implementation of the frameworks defined by the working group. Hyperledger Fabric runs a voting based algorithm, Hyperledger Indy runs Redundant Byzantine Fault Tolerance (RBFT), Hyperledger Iroha runs the Sumeragi algorithm (permissioned server reputation system) and Hyperledger Sawtooth runs the Proof of Elapsed Time (PoET) algorithm as defined by the Working Group Community to enable different frameworks that are enabled for different use cases.

Hyperledger smart contracts

Chaincode is a useful function in the Hyperledger platform, and helps to manage the state of the transaction and its associated business processing functionalities. Therefore, it typically acts as a 'smart contract' in the context of the Hyperledger platform, and it can be used to query or update transactions in the platform. With this facility, Hyperledger provides a modular architecture, can be deployed using container architecture and hence offers the advantage of scalable and high-performance architecture.

Chaincode is the central implementation of how the transaction is to be handled and managed by the platform owner. It is a small piece of code written as functions on how to handle the transactions, including the logic to execute, as well as the processing states of the transaction and the application handling capabilities, including the framework to be invoked through the API layer.

Hyperledger Fabric

Hyperledger Fabric is a modular blockchain framework that is the *de facto* standard for enterprise blockchain platforms delivering a high degree of confidentiality, resilience, scalability and flexibility. It was actually proposed to support varied pluggable implementations of different components and accommodate the complexity that exists across the entire economic ecosystem.

Hyperledger Fabric doesn't require expensive mining computations to commit transactions; so blockchains can scale up with low latency. Hyperledger Fabric is different from other blockchains like Ethereum or Bitcoin, not only because of its type, but also its internal functional mechanism.

Hyperledger Fabric is based on the following key design features.

- **Chaincode:** Programs are written in a higher level language that executes against the ledger's current state database.
- **Channels:** These are communication subnets for sharing information among members in the network in a highly confidential manner.
- **Endorsers:** Used for validating the transactions.
- **Membership services providers:** Provide identity validations and authentication processes by issuing and validating certificates.

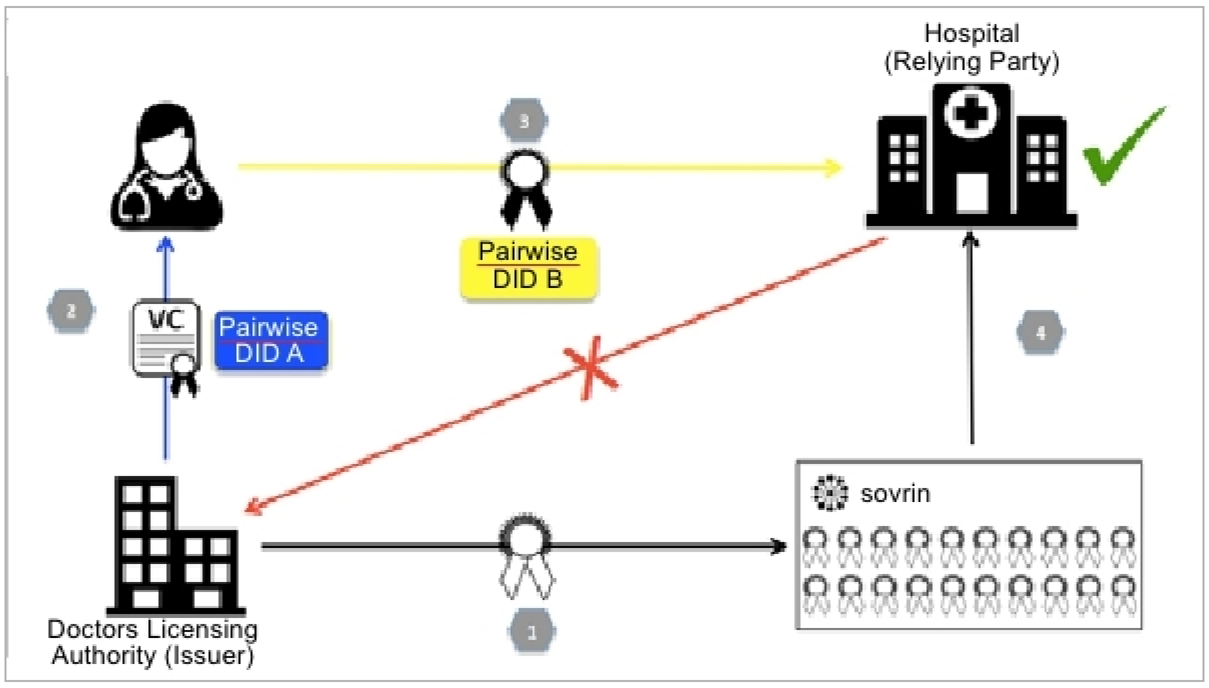

Figure 5: Hyperledger Indy architecture